



Fourth Edition

Photovoltaic Systems Engineering

Roger Messenger
Amir Abtahi



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**Roger Messenger and
Homayoon “Amir” Abtahi**



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We did not inherit this world from our parents. . . .

We are borrowing it from our children.

(Author unknown)

*It is our fervent hope that the engineers
who read this book
will dedicate themselves to the creation of a world
where children and grandchildren will be left
with air they can breathe and water they can drink,
where humans and the rest of nature
will nurture one another.*



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Preface

The goal of this textbook is to present a comprehensive engineering basis for photovoltaic (PV) system design so that the engineer can understand the *what*, the *why*, and the *how* associated with the electrical, mechanical, economic, and aesthetic aspects of PV system design. This book is intended to *educate* the engineer in the design of PV systems so that when engineering judgment is needed, the engineer will be able to make intelligent decisions based upon a clear understanding of the parameters involved. This goal differentiates this textbook from the many design and installation manuals that are currently available that *train* the reader *how* to do it but not *why*.

Widespread acceptance of the first three editions, coupled with significant growth and new ideas in the PV industry, along with additional years of experience with PV system design and installation for the authors, and, for that matter, a bit of nudging from the publisher, have led to the publication of this Fourth Edition. In recent years, annual installed renewable capacity has exceeded annual installed nonrenewable capacity in the United States. So who knows, there might even be jobs waiting out there for many who learn the material in this book.

The *what* question is addressed in the first three chapters, which present an updated background of energy production and consumption, some mathematical background for understanding energy supply and demand, a summary of the solar spectrum, how to locate the sun, and how to optimize the capture of its energy, as well as the various components that are used in PV systems. [Chapter 3](#) has been shortened a bit so actual design work can begin by about the fifth week of a three-credit course, but it also introduces lithium and a few other emerging battery technologies.

The *why* and *how* questions are dealt with in the remaining chapters in which every effort is made to explain why certain PV designs are done in certain ways as well as how the design process is implemented. Included in the *why* part of the PV design criteria are economic and environmental issues that are discussed in [Chapters 8](#) and [9](#). [Chapter 5](#) has been embellished with additional practical considerations added to the theoretical background associated with mechanical and structural design.

[Chapters 4](#), [6](#), and [7](#) have once again have been updated, including updated homework problems, to incorporate the most recently available technology and design and installation practice. In particular, [Chapter 4](#) incorporates more emphasis on higher voltage systems, new developments in IEEE 1547, new fire code requirements, and *National Electrical Code* changes in the 2014 Edition, such as rapid shut-down and arc fault protection. New sections on Arc Flash Calculations and System Commissioning have been added to [Chapter 4](#).

By the end of [Chapter 4](#), instructors can assign relevant design problems earlier in the course to avoid having designs due at final exam time. We have found when significant design projects are due just before final exams, students tend to spend time on design projects that might better be spent studying for the final exam, especially if

the final exam counts more toward the overall course grade. Paradoxically, the better designers have sometimes ended up with lower course grades.

Since the publication of the previous edition in 2009, a nearly overwhelming wealth of new research on old and new technologies has been underway. We try to cover the highlights of some of this activity in [Chapter 11](#) but have eliminated some of the historical content to make room for the new stuff.

The Appendix presents a recommended format for submittal of a PV design package for permitting or for design review that has been widely used in Florida by the authors and their associates.

A modified top-down approach is used in the presentation of the material. The material is organized to present a relatively quick exposure to all of the building blocks of PV systems followed by design, design, and design. Even the physics of PV cells of [Chapter 10](#) and the material on emerging technologies of [Chapter 11](#) are presented with a design flavor. The focus is on adjusting the parameters of PV cells to optimize their performance as well as on presenting the physical basis of PV cell operation.

Homework problems are incorporated that require both analysis and design, since the ability to perform analysis is the precursor to being able to understand how to implement good design. Many of the problems have multiple answers, such as “Calculate the number of daylight hours on the day you were born in the city of your birth.” We have eliminated a few homework problems based on old technology and added a number of new problems based on contemporary technology. Hopefully there are a sufficient number to enable students to test their understanding of the material.

We recommend the course be presented so that by the end of [Chapter 4](#) students will be able to engage in a comprehensive, relevant design project and by the end of [Chapter 7](#) they will be able to design relatively complex systems. We like to assign two design projects: a straight grid-connected system based on [Chapter 4](#) material and a battery-backup grid-connected system based on [Chapter 6](#) material. At the discretion of the instructor, an additional design of a stand-alone system might be considered at the end of [Chapter 7](#), or the stand-alone design might be assigned rather than the battery-backup grid-connected system.

While it is possible to cover all the material in this textbook in a three-credit semester course, it may be necessary to skim over some of the topics. This is where the discretion of the instructor enters the picture. For example, each of the design examples of [Chapter 4](#) introduces something new, but a few examples might be left as exercises for the reader with a preface by the instructor as to what is new in the example. Alternatively, by summarizing the old material in each example and then focusing on the new material, the *why* of the new concepts can be emphasized.

The order of presentation of the material actually seems to foster a genuine reader interest in the relevance and importance of the material. Subject matter covers a wide range of topics, from chemistry to circuit analysis to electronics, solid-state device theory and economics. *The material is presented at a level that can best be understood by those who have reached the upper division at the engineering undergraduate level and have also completed the coursework in circuits and in electronics.*

We recognize that the movement to reduce credit toward the bachelor's degree has left many programs with less flexibility in the selection of undergraduate elective courses, but note the material in this textbook can also be used for a beginning graduate level course.

While the primary purpose of this material is for classroom use, with an emphasis on the electrical components of PV systems, we have endeavored to present the material in a manner sufficiently comprehensive that it will also serve practicing engineers as a useful reference book.

The course can be successfully taught as an Internet course with a preference for live participation open to all. Those remote students who were sufficiently motivated to keep up with the course generally reported that they found the text to be very readable and a reasonable replacement for lectures. We highly recommend that if the Internet is tried that quizzes be given frequently to coerce students into feeling that this course is just as important as their linear systems analysis course. Informal discussion sessions can also be useful in this regard.

The PV field is evolving rapidly. While every effort has been made to present contemporary material in this work, the fact that it has evolved over a period of a year almost guarantees that by the time it is adopted, some of the material will be outdated. For engineers who wish to remain current in the field, many of the references and websites listed will keep them up to date. Proceedings of the many PV conferences, symposia, and workshops, along with manufacturers' data, are especially helpful.

This textbook should provide engineers with the intellectual tools needed for understanding new technologies and new ideas in this rapidly emerging field. The authors hope that at least one in every 4.6837 students will make their own contribution to the PV knowledge pool.

We apologize at the outset for the occasional presentation of information that may be considered to be practical or, perhaps, even interesting or useful. We fully recognize that engineering students expect the material in engineering courses to be of a highly theoretical nature with little apparent practical application. We have made every effort to incorporate heavy theory to satisfy this appetite whenever possible.

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Disclaimer

It's been said that no two snowflakes are the same. And, for that matter, it is unlikely that the weather on any 2 days at any location will be exactly the same. The minute-by-minute temperature and cloud cover will differ between any 2 days, whether they are one after the other or the same date on two successive or nonsuccessive years. As a result, estimating the performance of a PV system is, at best, an inexact science, since performance is critically weather dependent.

However, the longer the period included in a prediction or simulation of PV system performance, the more reliable the prediction becomes, since weather conditions tend to average out over the long term. Still, predicting the future is not nearly as reliable as recording the past. As a result, *when system performance is simulated in this text* using the System Advisory Model (SAM) developed by the U.S. National Renewable Energy Laboratory, because of the careful modeling tools used by NREL, SAM is one of the most reasonable tools to use to estimate annual energy production for a PV system. But no claim has been made that SAM is exact or perfect, and thus when SAM is used to estimate system performance, it must be done with the understanding of the following disclaimer:

“USER AGREES TO INDEMNIFY DOE/NREL/ALLIANCE AND ITS SUBSIDIARIES, AFFILIATES, OFFICERS, AGENTS, AND EMPLOYEES AGAINST ANY CLAIM OR DEMAND, INCLUDING REASONABLE ATTORNEYS’ FEES, RELATED TO USER’S USE OF THE DATA. THE DATA ARE PROVIDED BY DOE/NREL/ALLIANCE “AS IS,” AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL DOE/NREL/ALLIANCE BE LIABLE FOR ANY SPECIAL, INDIRECT OR CONSEQUENTIAL DAMAGES OR ANY DAMAGES WHATSOEVER, INCLUDING BUT NOT LIMITED TO CLAIMS ASSOCIATED WITH THE LOSS OF DATA OR PROFITS, THAT MAY RESULT FROM AN ACTION IN CONTRACT, NEGLIGENCE OR OTHER TORTIOUS CLAIM THAT ARISES OUT OF OR IN CONNECTION WITH THE ACCESS, USE OR PERFORMANCE OF THE DATA.”

In other words, using SAM for educational purposes is just fine, but blaming SAM if a system is built and does not work as predicted by SAM is not okay.



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Acknowledgments

Thanks to a long list of PV pioneers for making this publication possible. Only a few years ago, PV was considered to be an emerging technology. Some thought it would never make it on a large scale. But others with more persistence have persevered over the years such that PV is no longer a hobby but a mature, widely accepted, large-scale, worldwide technology that is growing faster than most advocates ever imagined.

One of these PV pioneers is Jerry Ventre, who coauthored the first three editions of this text. Jerry has been retired for a while now and has spent much of his time doing volunteer work with organizations that focus on PV job growth. It has been an honor and a privilege to work with Jerry over the years, and we wish him the best in his current and future activities.

We are convinced that it is virtually impossible to undertake and complete a project such as this without the encouragement, guidance, and assistance from a host of friends, family, and colleagues.

For this edition, we again asked many questions of many people as we rounded up information for the wide range of topics contained herein. A wealth of information flowed our way from the National Renewable Energy Laboratory (NREL) as well as from many manufacturers and distributors of a diverse range of PV system components. We feel confident that had we asked, we would have been able to obtain additional cooperation and materials from many more.

Special thanks to Kimandy Lawrence of VB Engineering for the many discussions during and after normal work hours on various nuances associated with certain systems, including, but not limited to, ac-coupled battery-backup systems, three-phase battery-backup PV systems, microinverter systems, and other experimental designs as well as the latest administrative and permitting issues. We would also like to thank Ms. Hadis Moradi, PhD Candidate, for her review and comments on the latest developments in microgrid systems.

Last, but not least, we thank Mimi Abtahi and Jane Caputi for encouraging us to work on this manuscript rather than play golf, tennis, watch TV, or pay attention to them.

Roger Messenger
Homayoon “Amir” Abtahi
August 2016



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In 1985, he installed the first solar-power system in Venezuela and was responsible for the first application of solar power for posthurricane emergency power and lighting and Ham radio communication operations in the aftermath of Hurricane Hugo in St. Croix in 1989 and Hurricane Marilyn in St. Thomas in 1995. In 1989, he published the first comprehensive catalog of 12-V appliances for use with PV systems.

Recently, he has been involved with PV installations in the Caribbean, South America, Bangladesh, and India. From 2008 to 2010, he was responsible for design

and installation of over 100 residential and 20 commercial/industrial PV systems. Over the past 15 years, he has had responsibility for the design and installation of 1 million BTUD of solar hot water and solar process heat. Along with PV and thermal applications, he has had experience with heat exchangers, MEP plan review, LEED projects, tracking PV, micro-turbines, parabolic trough solar, and other hybrid applications.

Abbreviations

AFCI	arc fault circuit interrupter
AGM	absorbent glass mat
AGS	automatic generator start
AM	air mass
ARC	anti-reflective coating
AWG	American Wire Gauge
BTU	British Thermal Unit
CEC	California Energy Commission
DH	daylight hours
DPP	differential power processor
EHP	electron-hole pair
EVA	ethylene vinyl acetate
GFDI	ground fault detection and interruption
LCB	linear current booster
MOSFET	metal oxide semiconductor field effects transistor
MPPT	maximum power point tracker
NEC	National Electrical Code
NFPA	National Fire Protection Association
NOCT	nominal operating cell temperature
Pa	In finance: Cumulative present worth factor
Pa	In engineering: Unit of pressure-Pascal
Pr	present worth factor
PW	present worth
PUC	point of utility connection
SFS	Sandia Frequency Shift
SOC	state of charge
STC	standard test conditions
SVS	Sandia Voltage Shift
THD	total harmonic distortion
UL	Underwriters Laboratory



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CHAPTER 1

Background

1.1 INTRODUCTION

On June 26, 1997, the U.S. Million Solar Roofs Initiative (MSRI) was announced at the United Nations Special Session on Environment and Development in New York [1]. It was proclaimed that “Now we will work with businesses and communities to use the sun’s energy to reduce our reliance on fossil fuels by installing solar panels on 1 million more roofs around our nation by 2010. Capturing the sun’s warmth can help us to turn down the Earth’s temperature.” In 1997, little concern was being expressed in the public sector about energy problems and perhaps even less discussion related to global warming, so this statement by President Clinton carries special significance from a 2016 perspective. Now, less than 20 years later as this chapter goes to press, both topics are receiving significant attention. By the end of 2007, it is estimated that more than 600,000 solar systems had been installed around the United States [2] and the State of California had introduced its own MSRI [3]. The 600,000 installations included photovoltaic (PV), solar water heating, and solar pool heating systems. In the 12 months of 2013, over 145,000 residential PV systems were installed in the United States, in addition to over 1 GW of PV installed by U.S. utilities [4].

But installations have not been limited to the United States. Germany, China, Japan, Spain, and many other countries have also had even more aggressive renewable energy programs [2,3]. In 2013, China and Japan were the largest markets for PV [4] and the United States was falling behind.

In the United States, more and more states are now adopting renewable portfolio standards (RPSs), which set goals for the percentage of the electrical energy mix to be provided by renewable sources by a certain date. Implementation of these goals will require engineers who understand the how and why of PV system design. RPSs, coupled with carefully devised incentive programs, significant reductions in installed system cost, and significant improvements in reliability have now provided the momentum for a sustained effort in the deployment of solar technologies well beyond the year 2030. In fact, the MSRI may need to be extended to a 100 Million Solar Roofs Initiative to meet the sustainable energy needs of future generations. According to *CNN Money*, the solar industry added 35,000 jobs in 2015 [5].

This book is dedicated to the engineers and technicians who have been and may become involved in turning this dream into reality.

1.2 POPULATION AND ENERGY DEMAND

The human population of the Earth has now passed 7.3 billion [6], and all of these inhabitants need the energy necessary to sustain their lives. Exactly how much energy is required to meet these needs and exactly what sources of energy will meet these needs will be questions to be addressed by present and future generations. One certainty, however, is that developing nations will be increasing their per capita energy use. For example, in 1997, the Peoples Republic of China was building electrical generating plants at the rate of 300 MW per week. These plants used relatively inexpensive, old, inefficient, coal-fired technology and provided electricity for predominantly inefficient end uses [7]. By 2008, pollution from factories and power plants in China had become so intense that some Olympic athletes had expressed reluctance to compete in the 2008 Olympic Games in Beijing for health reasons. The potential consequences to the planet of continued increase in the use of fossil fuels are profound [8,9]. Before proceeding with the details of PV power systems, a promising source of energy for the future, it is instructive to look at the current technical and economic energy picture. This will hopefully enable the reader to better assess the contributions that engineers will need to make toward a sustainable energy future for the planet.

1.3 CURRENT WORLD ENERGY USE PATTERNS

Figure 1.1 shows the increase in worldwide energy production by source between 1989 and 2014 [10]. In 2000, worldwide annual primary energy consumption was 397.40 quads [11]. The developed countries of the world consumed approximately 75% of this energy, while nearly 2 billion people in developing countries, mostly within the tropics, remained without electricity. In 2005, worldwide primary energy consumption increased to 462 quads [12] and 2 billion people still lived without electricity.

There can be a time delay between market forces and market responses. Note that the production of crude oil continued upward after the 1973 oil embargo and the subsequent significant crude oil price increases during the remaining 1970s and early 1980s. During this period, high crude oil prices spurred the development of energy efficiency legislation, such as the National Energy Conservation and Policy Act, codes for energy efficiency in building construction and increased vehicle fleet mileage requirements. Consumers also responded by reducing energy use by lowering thermostats and installing insulation and other energy conservation measures. The result was lower crude oil production for a period in the mid-1980s, since the demand was lower. During this same period, more efficient use of electricity resulted in the cancellation of nuclear plant construction, resulting in a significant decrease

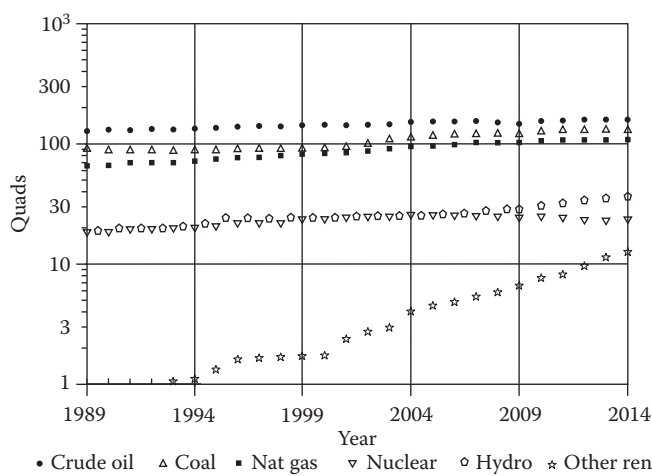


Figure 1.1 Growth of worldwide energy production by source. (Data from BP Statistical Review of World Energy, 64th Ed., June 2015, <http://www.bp.com/content/dam/bp/pdf/energy-economics/statistical-review-2015/bp-statistical-review-of-world-energy-2015-full-report.pdf>.)

in the growth rate of nuclear-produced electricity. Finally, concern over oil price control and embargoes prompted a switch from crude oil to coal and natural gas for use in fossil-fired electrical generation. Figure 1.2 shows the global mix of energy sources in the year 2013 [13]. The “renewables” category includes sources such as hydroelectric, wind, biomass, geothermal, and PV.

Figures 1.3 [13] and 1.4 [14] illustrate that the world faces a challenge of mammoth proportions as developing countries strive to achieve energy equity with

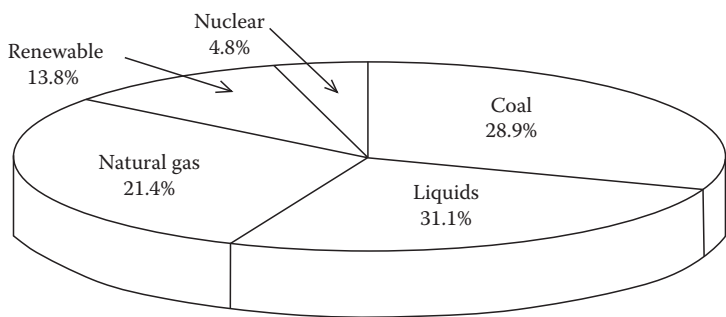
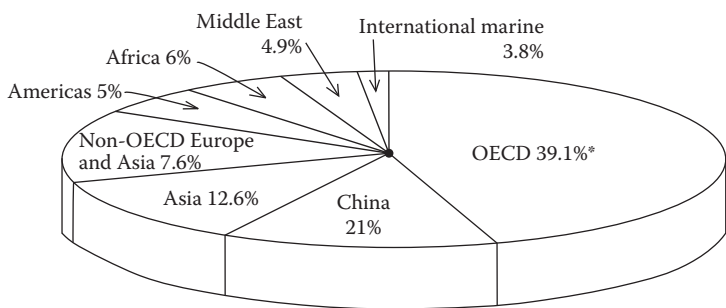


Figure 1.2 2013 global energy production mix by fuel type. (Data from International Energy Agency, 2015 key world energy statistics, 2013 total primary energy supply by source, http://www.iea.org/publications/freepublications/publication/KeyWorld_Statistics_2015.pdf.)



*Organization for economic cooperation and development

Figure 1.3 2013 distribution of energy users. (Data from International Energy Agency, 2015 key world energy statistics, 2013 total primary energy supply by source, http://www.iea.org/publications/freepublications/publication/KeyWorld_Statistics_2015.pdf.)

the developed countries. Note that energy equity is simply another term for the attempt to achieve comparable standards of living. But achieving a higher standard of living can carry with it a price. The price includes not only monetary obligations, but also the potential for significant environmental degradation if energy equity is pursued via the least expensive, first-cost options. Regrettably, this is the most probable scenario, since it is already underway in regions such as Eastern Europe and Asia. In fact, it is probably more likely that use of least-cost energy options may lead to comparable per capita energy use, but may simultaneously degrade the standard of living by producing air not suitable for breathing and

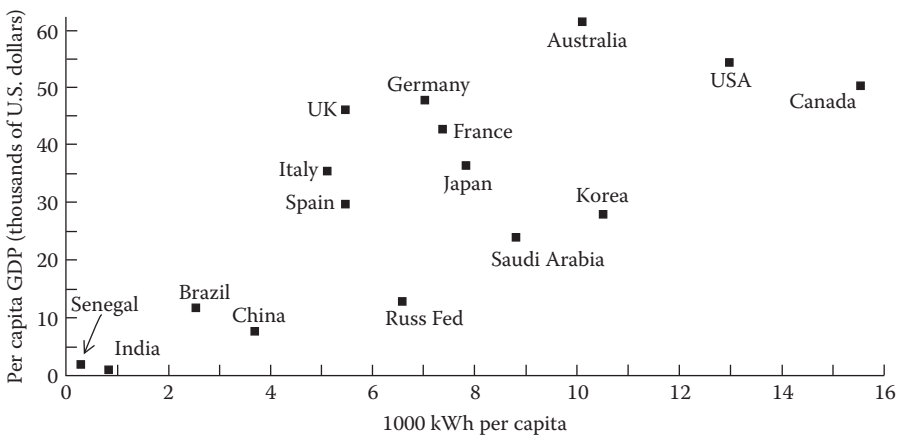


Figure 1.4 2013 worldwide per capita GDP versus per capita kW. (Data from The World Bank Data: Electric Power Consumption (kWh per Capita) and per Capita GDP, 2013, <http://data.worldbank.org/indicator/EG.USE.ELEC.KH.PC>.)

water not suitable for drinking. These issues will be dealt with in more detail in [Chapter 9](#).

What is missing in [Figure 1.3](#) is the efficiency with which the energy is consumed in these regions. [Figure 1.4](#) is particularly interesting because it clearly shows examples of countries where energy is used least and most efficiently [14]. Those countries with the highest ratios of per capita GDP to per capita kWh have developed more energy-efficient economies than those with low ratios. Some have shifted from predominantly manufacturing economies to predominantly information economies, with the result of a smaller fraction of energy use for manufacturing. Countries at the low end of the scale tend to be farming-oriented with the use of mostly manual labor or manufacturing oriented with very low wages of workers and low first-cost energy sources.

As developing countries increase their manufacturing capabilities, using cheap but polluting local energy sources, there may be pressure to relax pollution control standards within the developed world in order to maintain the ability to compete with production from developing countries. Opposition to free-trade agreements has been partially based on such environmental concerns, since such agreements forbid any country to impose import tariffs on goods produced in countries having weak environmental standards. So the developed countries are challenged to export efficiency to the developing countries. In fact, one of the arguments the United States made against the signing of the Kyoto Treaty on the reduction of greenhouse gases is that it does not require equal restrictions on greenhouse gas emissions in developing countries. The reasoning here is that it puts the United States at an economic disadvantage, but in [Chapter 9](#), the argument will be made that efficient use of energy, with minimal environmental degradation, leads to economic advantage. As of 2016, however, the situation has changed, with the United States and China both strongly supporting the Paris climate agreement of November–December 2015. Both countries, along with nearly 200 others [15], appear to have concluded that energy efficiency is good for business and the economy as well as for the environment.

But what does this discussion have to do with PV power production? Simple. As will be shown in [Chapter 9](#), PV energy sources are very clean, but current PV deployment costs have only recently become competitive with the initial installed direct costs of fossil sources of electrical generation in most cases. It means that the consumer must be familiar with life-cycle costing and that the engineer must be able to create the most cost-effective PV solution. It also means that a significant amount of research and development must be done to ensure the continuation of the decrease in the price of PV generation. It also means that work must continue in the effort to put a price tag on environmental degradation caused by energy sources, so this price can be factored into the total cost to society of any energy source.

The bottom line is that there remains a significant amount of work in research, development, and public education to be done in the energy field, and particularly in the field of PV. And much of this work can best be done by knowledgeable engineers who, for example, among other things, understand the concept of exponential growth.

1.4 EXPONENTIAL GROWTH

1.4.1 Introduction

Exponential growth is probably most familiar to the electrical engineer in the diode equation, in which I and V are the diode current and voltage, respectively, I_o is the reverse saturation current and kT/q is 26 mV at $T = 300$ K, or, specifically,

$$I = I_o(e^{(qV/kT)} - 1). \quad (1.1)$$

While this equation is fundamental to the performance of PV cells, many other physical processes are also characterized by exponential growth.

Exponential growth is commonly referred to as compound interest. Almost everyone has heard about it, but few understand the ramifications of constant annual percentage increase in a quantity, whether it be money, population, or energy supply or demand. One of the first to warn of the dangers of exponential growth was Malthus in 1798 [16]. He warned that population growth would exceed the ability to feed the population. The Malthusian theory is often the subject of ridicule of growth enthusiasts [17]. The intent of this discussion is neither to support nor to discount the predictions of Malthus, but merely to illustrate an important mathematical principle that engineers often overlook. The application of the principles of exponential growth is widespread in society, so the principles of exponential growth should be just as important to a well-informed engineer as is the second law of thermodynamics. For those who may have missed the second law of thermodynamics in either chemistry or thermodynamics class, it is a statement that in every process less energy comes out than what is put in. In other words, there is no free lunch.

1.4.2 Compound Interest

Compound interest is the process of compounding simple interest. If a quantity N_o is subject to an interest rate i , with i expressed as a fraction (i.e., $i = \%/100$), the quantity will increase (or decrease, if $i < 0$) to a value of

$$N(1) = N_o(1 + i) \quad (1.2)$$

after one prescribed time period has elapsed. If the quantity present after the prescribed time period is allowed to remain and to continue to accumulate at the same rate, then the quantity is subject to compound interest and the amount present after n time periods will be

$$N(n) = N_o(1 + i)^n. \quad (1.3)$$

To show that this formula is a form of the exponential function, one need only to recall that

$$y^x = e^{x \ln y}.$$

Hence,

$$N(n) = N_0 e^{n \ln(1+i)}. \quad (1.4)$$

Some special properties of the exponential function can now be considered.

1.4.3 Doubling Time

To determine the time, D , it will take for the original quantity to double, one need only to set $N(n) = 2N_0$ and solve for n . The result (and the answer to problem 1.1) is

$$n = \frac{\ln 2}{\ln(1+i)} = D. \quad (1.5)$$

For small values of i , $\ln(1+i)$ can be approximated as $\ln(1+i) \cong i$. Noting that $\ln 2 = 0.693$ leads to the formula so popular in the financial world, that is,

$$D \cong \frac{0.7}{i}. \quad (1.6)$$

Hence, for an interest rate of 7% per year ($i = 0.07$), the doubling time will be 10 years. For an interest rate of 10%, the doubling time will be approximately 7 years. However, as the interest rate exceeds 10%, the approximation becomes less valid, and the exact expression should be used for accurate results. In the case where the interest rate is negative, it should be obvious that no doubling can occur. The authors have proven this to be the case with various investments in the stock market.

An important property of the exponential function is that the doubling process continues for all time. Hence, if the doubling time is 10 years, then the quantity will double again in another 10 years, so it will now be 4 times its original value. In another 10 years it will double again to 8 times its original value. After 40 years, the quantity will be 16 times its original value. [Figure 1.5](#) shows this exponential increase.

Note that if the function $y = Ae^{bx} = A10^{bx \log e}$ is plotted with linear coordinates, the familiar exponential curve appears, as in [Figure 1.5a](#). If the logarithm of each side is taken, the result is

$$\log y = \log A + (b \log e)x. \quad (1.7)$$

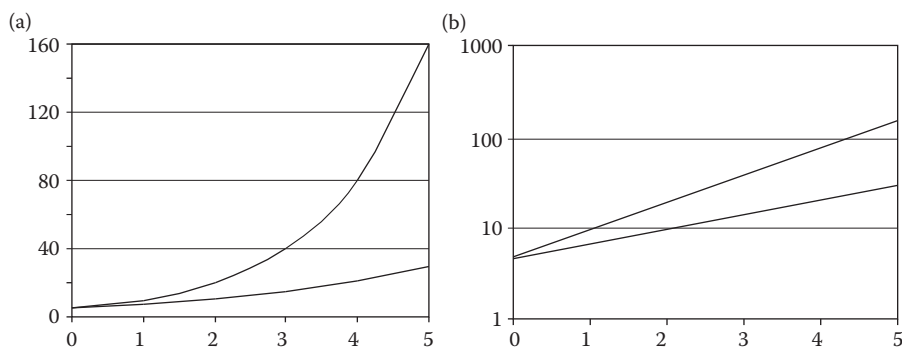


Figure 1.5 Examples of exponential functions plotted on linear and on semilogarithmic coordinates. (a) Linear vertical axis. (b) Logarithmic vertical axis.

Hence, if $\log y$ is plotted as a function of x , the graph will be linear with slope $b \log e$, as shown in [Figure 1.5b](#). [Figure 1.5b](#) shows that plotting $\log y$ versus x is a convenient way to check for an exponential relationship between two variables. Notice also that the vertical axis can be labeled either in terms of $\log y$ or, simply, in terms of y . When the axis is labeled in terms of y , it is understood that the axis is linear in the logarithm of y . For the person who prefers to let computers do the work, either Excel or MATLAB® can conveniently plot data and produce a least mean square curve fit to the data along with an estimate of the goodness of the fit. When a set of data is available, all one needs to do is to list the independent and dependent variables side-by-side on an Excel spreadsheet and then use the Chart Wizard to plot an x - y scatter plot of the data. By selecting of the chart option, and then selecting the plot that shows data but no connecting lines, one can then click on one of the data points on the chart, which will select the data set on the graph. Then go to “Chart” on the pull-down menu, and select “Add Trendline.” This option opens a dialog box that offers a choice of six trend/regression types. To obtain the best fit, select a regression type, then select “Options” and choose to add the equation and the R^2 value to the graph. An R^2 value of 1 indicates a perfect fit of the data to the curve by the equation that is displayed. Low R^2 values suggest choosing a different regression type. Several good user guides for Excel are available for those who would like to further explore the use of Excel as a convenient analysis tool for use in PV system design or other technical endeavors [18,19]. MATLAB and other math programs are also powerful tools.

For the reader who has not had the pleasure of using Excel to find the least mean square fit to a data set, several problems are available at the end of this chapter. It is anticipated that the reader will use Excel frequently as a tool for the design of PV systems in later chapters.

1.4.4 Accumulation

Another important property of the exponential function is the amount of accumulation (or depletion) of a quantity in a doubling time. Noting that integration of

a function yields the area of the curve under the function, which is the accumulation of the quantity being integrated, it is a straightforward math exercise to integrate Equation 1.4 to show that *the amount accumulated (or depleted) in a doubling time is equal to the amount accumulated (or depleted) in all previous history* (see Problem 1.1).

The implications of this result are far reaching. For example, a prominent political figure once noted that there was still as much oil underground in the United States as what had been pumped out since pumping first started more than 140 years ago. This was at a time when oil extraction was increasing at a rate of approximately 7% per year. If the extraction had continued to increase at 7% per year, which has a doubling time of approximately 10 years, in the next doubling time all of the remaining petroleum would have been extracted. Many other important public figures have made similar statements that tend to assign a linear nature to the exponential function [20]. Could this be an argument for engineers to run for public office?

In fact, extraction did not continue to increase at this rate. With regard to the use of resources, Hubbert [21] developed a model that incorporates a Gaussian function for depletion, which seems to have more validity than the exponential model. The rising edge of the Gaussian function, however, is conveniently approximated by an exponential.

The accumulation formula, of course, may also apply to the deployment of new technology. Since the early 1990s, PV power production has been increasing at a very impressive rate. Problems 1.15 and 1.16 offer an opportunity to explore the relevance of this observation if this rate of increase continues.

1.4.5 Resource Lifetime in an Exponential Environment

The previous discussion of exponential growth has been based on total amounts of a quantity at any given time. If the time derivative of the exponential expression for quantity is taken, the rate of change of the quantity is obtained. Since the derivative of an exponential is also an exponential, the same rules apply to the derivative as to the function. Distinguishing between amount present and rate of use (or increase) is important when determining the lifetime of a resource. Hence, when considering an exponential expression, one needs to establish whether it refers to barrels or barrels per day, or, perhaps, megawatts or megawatts per year of PV deployment.

The final concept to explore relating to the exponential function is the lifetime of a resource under conditions of exponential increase. It is common to predict the lifetime of a resource under the current rate of consumption. This involves simple arithmetic, since if there are Z (quantity) widgets left to use and if we use X (rate of use) of them per year, then the widgets will last for Y years, where $Y = Z/X$. But what happens to the expected lifetime of the widget if people decide that they really like widgets and they decide to use them at an increasing rate of 100i% per year, noting that if i represents a fraction, then $100i$ converts the fraction to a percentage? This problem can be solved by assuming that C_0 represents the present rate of consumption of a resource and Y_0 represents the estimated lifetime of the resource

at the present rate of consumption. Then, if consumption increases by $100i\%$ per year, the rate of consumption at any point in time, x , is given by

$$C(x) = C_o(1 + i)^x. \quad (1.8)$$

The total accumulated consumption over a period of m years, TOT, can be found from previous formulas, or, more formally, by evaluating

$$\begin{aligned} \text{TOT} &= \int_0^m C(x) dx = \int_0^m C_o e^{x \ln(1+i)} dx \\ &= \frac{C_o}{\ln(1+i)} (e^{m \ln(1+i)} - 1). \end{aligned}$$

Next, set the total to equal the estimated amount remaining ($C_o Y_o$) and solve for m , since this will yield the number of years for the total consumption to equal the amount remaining. The result is

$$m = \frac{\ln[Y_o \ln(1+i) + 1]}{\ln(1+i)}. \quad (1.9)$$

As an example of the use of this result, assume that a resource is estimated to last for another 100 years at present consumption rates ($Y_o = 100$), but that consumption will increase at the rate of 5% per year ($i = 0.05$). Using these numbers in the above formula gives $m = 36.31$ years. If the estimate is off by a factor of 10, and there is really a 1000-year supply left at current consumption rates, then $m = 80.09$ years.

As perhaps a more reassuring example of the use of this result, it is also possible that the consumption of a resource might decline at a constant percentage per year. This could happen if the resource were to be replaced by another resource, for example. For the above example, with $Y_o = 100$ years and an annual decrease of 0.5% (i.e., $i = -0.005$), the new lifetime becomes 139 years, and if $i = -0.01$, the resource will last forever.

Hence, two important observations emerge from the lifetime formula:

1. If annual consumption of a resource increases exponentially, it is not important how much is thought to remain; it will be consumed much faster than one can imagine.
2. If annual consumption decreases exponentially, it is possible to extend the lifetime of a resource to forever.

1.5 HUBBERT'S GAUSSIAN MODEL

In 1956, M. King Hubbert, who was employed by the Shell Oil Company, published his now acclaimed theory of resource depletion [21]. Simply put, Hubbert

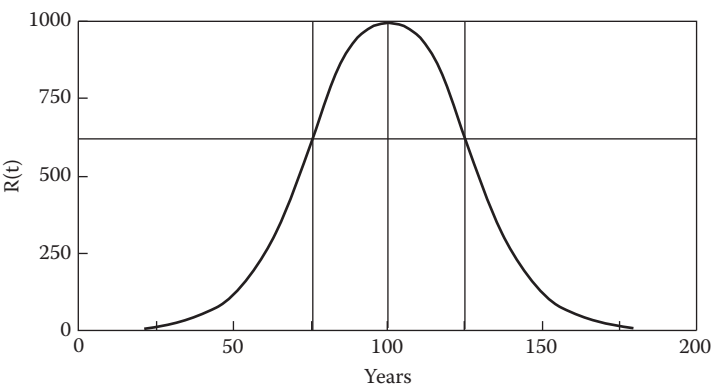


Figure 1.6 Gaussian function with $R_m = 1000$, $t_0 = 100$, and $s = 25$.

reasoned that the life of a finite resource follows a Gaussian curve described by the equation, often referred to as the error function or normal curve,

$$R(t) = R_m e^{-(t-t_0)^2/2s^2}, \tag{1.10}$$

where $R(t)$ represents the consumption rate at a given time, t , R_m represents the maximum consumption rate, and s represents a shape factor for the curve, commonly known as the standard deviation.

He also observed that production of petroleum lagged discovery by approximately 40 years. [Figure 1.6](#) shows a plot of Equation 1.10, and [Figure 1.7](#) [21] shows

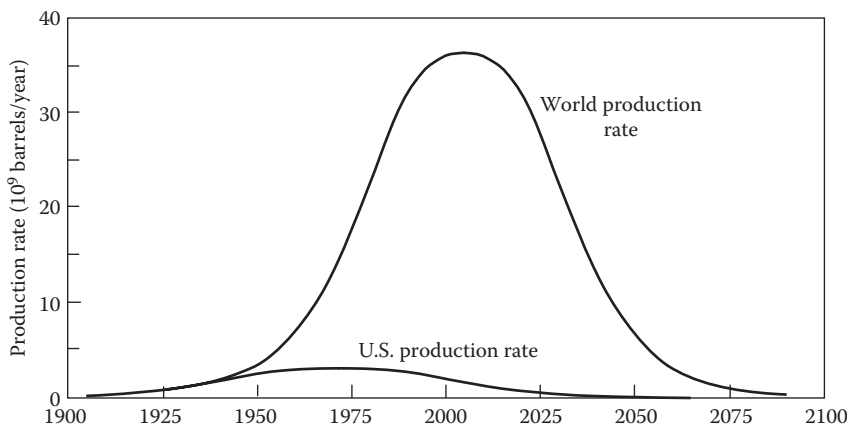


Figure 1.7 Hubbert's predictions for U.S. and world petroleum production. (Adapted from Hubbert, M. K., *American Petroleum Institute, Drilling and Production Practice*, 23, 1956, 7–25, with permission.)

Hubbert's 1956 curves relating to petroleum production. The rising edge represents a nearly exponential function, until it nears the peak of the curve, where leveling occurs, followed by nearly exponential decay. Since the curve of [Figure 1.6](#) plots consumption rate, note that according to Hubbert's theory, when half a resource is consumed, the consumption will have leveled at its maximum value and then will begin its nearly exponential decline. According to Hubbert's model, domestic petroleum production in the United States would peak near 1970 and world petroleum production would peak near the turn of the century. Indeed, as 2008 approached, it appeared that Hubbert may have been correct. According to projections of the U.S. Department of Energy, Energy Information Agency, both domestic and Alaskan petroleum production would continue to decline through the year 2015 [\[22\]](#). Current data, however, show healthy U.S. domestic production as a result of enhanced hydraulic fracturing and horizontal drilling (fracking) of petroleum and natural gas, a glitch that was not anticipated in 1956.

Interestingly enough, it also turns out with the Hubbert theory that if an error is made in estimating the amount of a resource present, the peak of the curve will not be shifted by a significant amount. Perhaps, the most significant conclusion of the Hubbert theory, however, is that the consumption of a resource follows a smooth curve rather than an abrupt one that involves unabated consumption until suddenly no more is left. Hopefully, approaching the peak sends a message to find a replacement. For that matter, exceeding 400 ppm of CO₂ in the atmosphere presents an even more compelling reason for a carbon replacement. In fact, the rapid increase in deployment of renewables as shown in [Figure 1.1](#) suggests that perhaps the replacement has already been found.

1.6 NET ENERGY, BTU ECONOMICS, AND THE TEST FOR SUSTAINABILITY

The net energy associated with an energy source is simply the difference between the energy required to obtain and convert the source to useful energy and the actual useful energy obtained from the source. For example, in order to be able to burn a barrel of oil, it is necessary to find the oil, extract the oil, transport the oil, refine the oil, and construct the facility for burning the oil. The refined oil must then be transported to the burning site and, presumably, after the oil is burned, any environmental damage resulting from the extraction, transportation, refining, and burning should be repaired. Energy is involved in all of these steps. The bottom line is that if it takes more than a barrel of oil worth of energy to obtain and convert the energy available in a barrel of oil, one should question seriously whether it makes sense to burn the oil in the first place.

In some cases, it may make sense to expend the energy to get the resource. Suppose, for example, that another use were discovered for oil, such as providing an essential chemical for the cure of cancer. Then it would make sense to expend energy from sources other than oil, even if the energy exceeded the energy value of the oil, in order to make the oil available for the more important use. Another situation

would be to use a lower grade or quality form of energy to produce a higher grade or quality form of energy. Sometimes such an action might make energy sense.

For example, burning coal to produce electricity takes about three units of coal energy to produce one unit of electrical energy. Until television sets that run directly from coal are invented, this inefficient process of converting coal energy to electrical energy will probably continue, in spite of the known environmental problems with coal.

The concept of net energy was introduced by Odum and Odum in 1976 [22,23]. They incorporated the net energy concept into a new standard for economics that they felt made better sense than the gold standard. They called it the BTU standard. The BTU standard simply recognizes that everything has an energy content. Henderson [24] has written extensively on the concept of BTU economics. The reader is encouraged to read Odum and Odum and Henderson during a term break for enlightening discussions of how the economic system might be changed to an energy-based standard.

For the purposes of this book, the test for sustainability for an energy source will include two factors. The first will be whether the source is finite. A finite source is generally termed nonrenewable, while an infinite source is termed renewable. The second will be whether the source has positive net energy. That is, once energy is expended to produce the source, will the source then generate more energy than was required for its production?

The idea that a source can generate more energy than was expended in its production may seem inconsistent with the second law of thermodynamics. However, if we allow the use of energy from a very large reservoir as a supply of energy to be converted by the source, then the source becomes nearly infinite. In the case of the sun, which is expected to survive for another 4 billion years or so [25], we have such a reservoir. Thus, for example, if a PV cell can generate more electrical energy over its lifetime than was expended in its production and deployment and ultimately in its disposal, including environmental energy costs, then the cell would be considered to have positive net energy.

The concept of net energy will be considered in the context of PV cell production and in the discussion of environmental effects of energy sources.

1.7 DIRECT CONVERSION OF SUNLIGHT TO ELECTRICITY WITH PV

Becquerel first discovered that sunlight can be converted directly into electricity in 1839, when he observed the photogalvanic effect [26]. Then, in 1876, Adams and Day found that selenium has PV properties. When Planck postulated the quantum nature of light in 1900, the door was opened for other scientists to build on this theory. It was in 1930 that Wilson proposed the quantum theory of solids, providing a theoretical linkage between the photon and the properties of solids. Ten years later, Mott and Schottky developed the theory of the solid-state diode, and in 1949, Bardeen, Brattain, and Shockley invented the bipolar transistor. This invention, of

course, revolutionized the world of solid-state devices. The first solar cell, developed by Chapin, Fuller, and Pearson, followed in 1954. It had an efficiency of 6%. Only 4 years later, the first solar cells were used on the Vanguard I orbiting satellite.

One might wonder why it took so long to develop the PV cell. The answer lies in the difficulty in producing sufficiently pure materials to obtain a reasonable level of cell efficiency. Prior to the development of the bipolar transistor and the advent of the space program, there was little incentive for preparing highly pure semiconductor materials. Coal and oil were meeting the world’s need for electricity, and vacuum tubes were meeting the needs of the electronics industry. But since vacuum tubes and conventional power sources were impractical for space use, solid state gained its foothold.

PV cells are made of semiconductor materials and are assembled into modules of 36 or more cells. This observation is significant, since this means the same industry that has, in the past 60 years, progressed from the development of the bipolar transistor to integrated circuits containing millions of transistors is also involved in the development of PV cells. Figure 1.8 [27] shows the decline in the cost of PV modules since 1977. Much of the initial cost reduction has been due to process improvement in the production of the cells. At this point, the limiting factor is becoming the energy cost of the cells. For a few years following 2005, refined silicon prices increased significantly due to a worldwide shortage, but then fell again in 2008 as supply caught up with demand. Hence, the challenge of the future will be to reduce the energy content of the cell production process while maintaining or increasing cell performance, efficiency, and reliability.

Figure 1.9 [28–32] shows world PV shipments in megawatts from 1975 to 2014. Note that the data are plotted on semilogarithmic coordinates. The actual data since 1995 are given in Problem 1.17, so the reader can generate a plot and determine the goodness of fit and the rate of growth over this period. A further important observation is that in 1995, 45% of the world’s PV modules were manufactured in the United

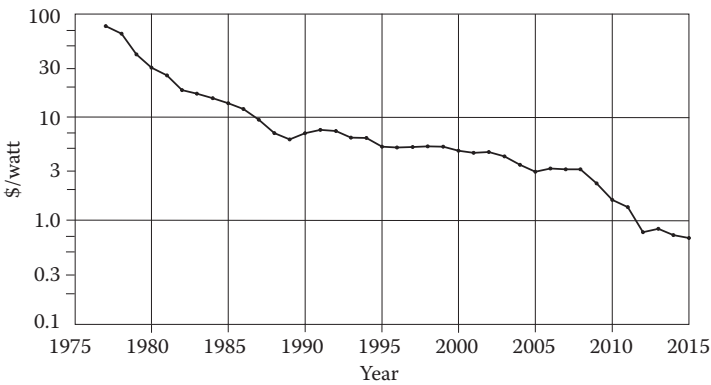


Figure 1.8 Decline in cost per watt for PV modules. (Data from Shahan, Z., 13 charts on solar panel cost and growth trends, Clean Technica, September 4, 2014, <http://cleantechnica.com/2014/09/04/solar-panel-cost-trends-10-charts/>.)

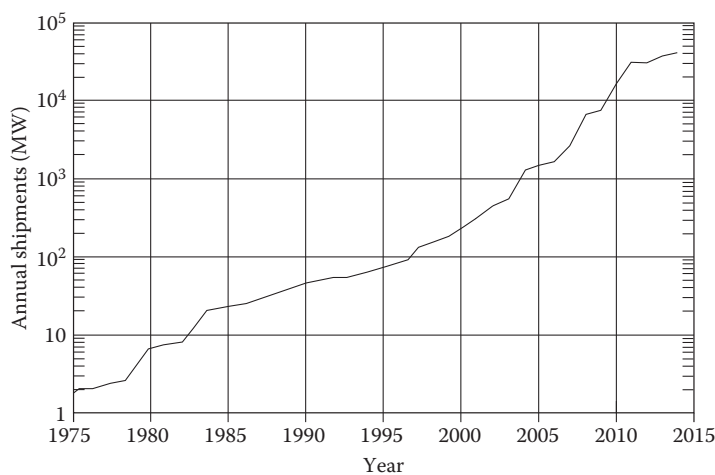


Figure 1.9 Worldwide PV shipments, 1975–2014. (Data from Growth of Photovoltaics, World wide, Wikipedia, https://en.wikipedia.org/wiki/Growth_of_photovoltaics#Deployment_by_country; *International Marketing Data and Statistics*, 22nd Ed., Euromonitor PIC, London, 1998; Maycock, P. D., The world PV market, production increases 36%, *Renewable Energy World*, July–August 2002, 147–161; Earth Policy Institute, Eco-economy indicators: SOLAR POWER—Data, www.earth-policy.org/Indicators/Solar/2007_data.htm; Maycock, P. D., *PV News*, 22(3), March 2003.)

States, while Europe, Japan, and the rest of the world manufactured 80% of the world’s PV modules in 2002 [30] and 93% of the world’s PV cells in 2007 [33]. Just as the United States has allowed the manufacture of consumer electronics to transfer to other countries, it appears that the United States is also allowing the same to happen with the PV industry. It will be interesting to observe this trend in the future.

As an initial test as to whether to continue with this book, it is useful to determine whether the net energy associated with PV cells is positive. Indeed, current PV cells can produce more than a 10:1 return on the energy invested in their production, and future improvements in production technology and practice may result in values up to 45:1 [34]. Hence, it appears to be worth investigating this technology in more detail. In order to compare various energy sources, it is helpful to compare the units with which the available energy in these sources is measured.

1.8 ENERGY UNITS

Energy is measured in a number of ways, including the calorie, the BTU, the quad, the foot-pound (ft-lb), and the kilowatt hour (kWh). When pounds (lb) and kilograms (kg) are used in this text, they will be treated as weight, rather than mass. Thus, a 220 lb person can also be described as a 100 kg person. For the benefit of those who may not have memorized the appendices of their freshman physics books,

we repeat the definitions of a number of these energy-related quantities for an Earth-based system at or about a temperature of 27°C [10,35].

- 1 calorie is the heat needed to raise the temperature of 1 g of water 1°C.
- 1 BTU is the heat needed to raise the temperature of 1 lb of water 1°F.
- 1 quad is 1 quadrillion (10^{15}) BTU.
- 1 ft-lb is the energy expended in raising 1 lb through a height of 1 ft.
- 1 kWh is the energy expended by 1 kW operating for 1 hour.
- 1 MTOE (million tons of oil equivalent) is the energy available from 1 million tons of crude oil.
- 1 Mbd (million barrels per day) is the energy available from 1 million barrels of crude oil.
- 1 Bcm (billion cubic meters) is the energy available from 1 billion cubic meters of natural gas.

With these definitions, and with the numerical values of MTOE, million barrels of oil equivalent (MBOE), and Bcm, the following equivalencies can be determined:

- 1 BTU = 252 calories
- 1 kWh = 3413 BTU = 2,655,000 ft-lb
- 1 ft-lb = 0.001285 BTU
- 1 quad = 10^{15} BTU = 2.930×10^{11} kWh
- 1 MTOE = 0.0397 quad
- 1 MBOE = 0.00541 quad
- 1 Bcm = 0.0357 quad

It should be noted that MTOE, MBOE, and Bcm represent the *total* amount of energy stored in the specified quantities of coal, oil, and natural gas. When these fuels are burned to produce electricity, the laws of thermodynamics limit the percentage of the available energy that can be converted to electricity to approximately 40%. In addition, some of the heat lost in the process is sometimes recovered for other use. So, in converting MTOE, for example, to kWh of useful electricity, an overall efficiency factor is needed for the process.

Because the emphasis of this text will be on electrical generation, and since the kWh is the common unit for electrical energy, the equivalence between kWh and ft-lb is especially noteworthy. For example, suppose a 150-lb person wished to generate 1 kWh, assuming a system with 100% efficiency. One way would be to climb to the top of a 17,700-ft mountain to create 1 kWh of potential energy. Then, by returning to sea level by way of a chair, connected via a pulley system to a generator, the person's potential energy could be converted to electrical energy. This kWh could then be sold at wholesale for about 6 cents. Another somewhat simpler method is to burn approximately 11 fluid ounces of petroleum to produce steam to turn a steam turbine as shown in [Figure 1.10](#).

Still another method is to deploy about 2 m² of PV cells. This system will produce about 1 kWh per day for 25 years or more with no stops for refueling, no noise, minimal maintenance, and no release of CO₂, SO₂, or NO₂ while the electricity is being produced.

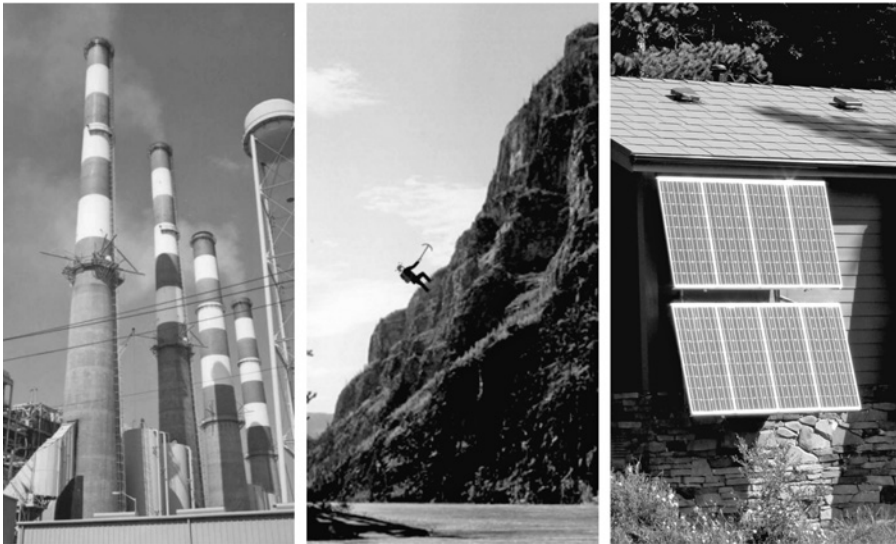


Figure 1.10 Several ways to produce a kWh of electricity.

PROBLEMS

- 1.1 Prove Equation 1.5.
- 1.2 Calculate the approximate and exact doubling times for annual percentage increases of 5%, 10%, 15%, and 20%.
- 1.3 The human population of the Earth is approximately 7 billion and is increasing at approximately 2% per year. The diameter of the Earth is approximately 8000 miles and the surface of the Earth is approximately two-thirds water. Calculate the population doubling time, then set up a spreadsheet that will show (a) the population, (b) the number of square feet of land area per person, and (c) the length of the side of a square that will produce the required area per person. Carry out the spreadsheet for 15 doubling times, assuming that the rate of population increase remains constant. What conclusions can you draw from this exercise?
- 1.4 Show that in an exponential growth scenario, the amount accumulated in a doubling time equals the amount accumulated in all previous history.
- 1.5 Assume there is enough coal left to last for another 300 years at current consumption rates.
 - a. Determine how long the coal will last if its use is increased at a rate of 5% per year.
 - b. If there is enough coal to last for 10,000 years at current consumption rates, then how long will it last if its use increases by 5% per year?
 - c. Can you predict any other possible consequences if coal burning increases at 5% per year for the short or long term?
 - d. Determine the annual percentage reduction in coal consumption to ensure that coal will last forever, assuming the 300-year lifetime at present consumption rates.

- 1.6 The half-life is the time it takes to decay exponentially to half the original amount. If the half-life of a radioactive isotope is 500 years, how many years will it take for an amount of the isotope to decay to 1% of its original value? Assume the isotope decays exponentially.
- 1.7 If a colony of bacteria lives in a jar and doubles in number every day, and it takes 30 days to fill the jar with bacteria,
- How long does it take for the jar to be half full?
 - How long before the bacteria notice they have a problem? (You may want to pretend you are a bacterium.)
 - If on the 30th day, three more jars are found, how much longer will the colony be able to continue to multiply at its present rate?
- 1.8 An enterprising young engineer enters an interesting salary agreement with an employer. She agrees to work for a penny the first day, 2 cents the second, 4 cents the third, and so on, each day doubling the amount of the previous day. Set up a spreadsheet that will show her daily and cumulative earnings for her first 30 days of employment.
- 1.9 Burning a gallon of petroleum produces approximately 25 lb of carbon dioxide and burning a ton of coal produces approximately 7000 lb of carbon dioxide.
- If a barrel of petroleum contains 42 gallons, if the world consumes 80 million barrels of petroleum per day and if the atmosphere weighs 14.7 lb per square inch of Earth surface area, calculate the weight of carbon dioxide generated each year from burning petroleum and compare this amount with the weight of the atmosphere.
 - If a total of 16 million tons of coal are burned every day on the Earth, calculate the weight of carbon dioxide generated each year from coal burning and compare it with the weight of the atmosphere.
- 1.10 Assume a world population of 7.3 billion and a U.S. population of 323 million.
- Look up the present total annual U.S. primary energy consumption. Then determine the total world energy consumption in quads if the rest of the world were to use the same per capita energy as in the United States
 - If the energy source mix were to remain the same as the present mix in achieving the scenario of Part a, what would be the percentage increase in CO_2 emissions?
- 1.11 Obtain data on worldwide energy consumption by sector from the U.S. Department of Energy, Energy Information Administration website. Plot the data and estimate annual percentage growth rates for the regions reported and then for the world.
- 1.12 The following measurements of $x(t)$ are made:

t	0	1	2.3	3.0	4.5	5.2	6.5	8.0
x	2.1	8.4	31	65	360	850	3700	20,000

Construct a semilog plot of $x(t)$ either manually or with a computer, and determine whether the function appears to have an exponential dependence. If so, determine $x(t)$.

- 1.13
- What does the area under the Gaussian curve represent?
 - Show that 68% of the area under the Gaussian curve lies within one standard deviation, s , of maximum value of the function.
 - What percentage of the area lies within $2s$?
- 1.14 Determine R_m , t_0 , and s for the worldwide graph of [Figure 1.7](#).

- 1.15 Look up actual U.S. and world petroleum production figures and plot them on Hubbert’s curves to compare the actual production with the theoretical production.
- 1.16 Based on the data of [Figure 1.9](#),

a. Estimate the year when PV shipments will reach 100 GW.

b. Estimate the year when PV shipments will reach 1000 GW.

c. Estimate the year when PV shipments will reach 2700 GW.
- 1.17 The Wikipedia [\[28\]](#) reports the following worldwide PV production figures. Plot the data on an Excel graph, establish an equation to represent the data, and then answer the three questions posed in Problem 1.16. Compare the results of the two problems.

Year	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
GW	0.57	1.08	1.39	1.55	2.52	6.66	7.34	17.2	30.1	30.0	38.4	40.1

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